

BDM refinement: validation and halo properties

Duplicate survival, consistent bound populations, and a
billion-particle test

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Fix halo identity and the bound population

Duplicate survival

- ▶ Equal-mass candidates can have slightly different particle sets.
- ▶ Remove exact member-set copies.
- ▶ Break mass ties by stable host priority; exclude inner centres.

Also: configure before peak finding; restore particles exactly; order corrected shapes as $0 \leq c/a \leq b/a \leq 1$.

Consistent properties

- ▶ Use the outermost supported SO crossing.
- ▶ Iterate unbinding until membership converges.
- ▶ Measure properties from the final bound set.

A billion-particle test on identical snapshots

Box / gravity	$512 h^{-1} \text{ Mpc}$; GR, flat ΛCDM
Particles / mesh	$1024^3 / 2048^3$
Particle mass	$1.07 \times 10^{10} h^{-1} M_{\odot}$
PM cell	$0.25 h^{-1} \text{ Mpc}$
Initial / saved redshifts	100 / 2, 1, 0

- Both finders use the same saved particles and requested virial convention.
- Published cut: $M_{\text{bound}} \geq 2.5 \times 10^{12} h^{-1} M_{\odot}$; $R_{\text{ext}} = 0.15$.

The published $z = 0$ abundance guide is $10^{12.5} h^{-1} M_{\odot}$. New convergence evidence requires a resolution series.

Ruan et al., MG-GLAM, arXiv:2110.00328, Appendix A.

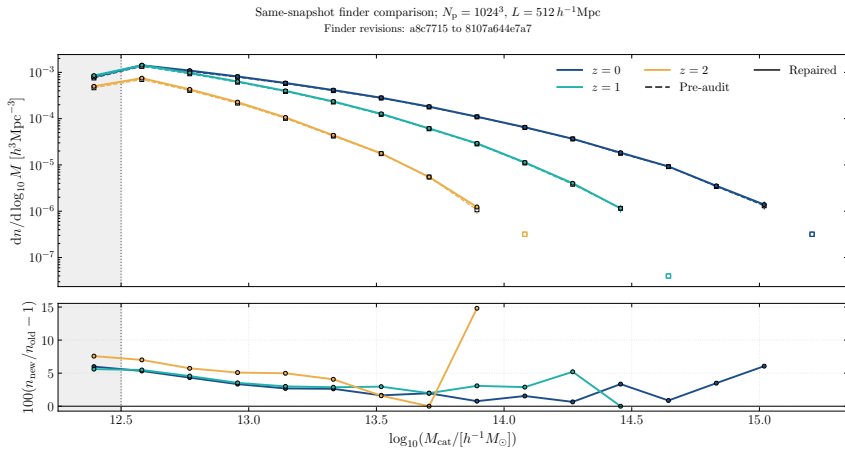
Membership and host checks pass

z	Pre-audit	Refined	Duplicate sets	Host violations
2	49,205	52,370	0	0
1	114,454	119,688	0	0
0	141,357	147,042	0	0

Duplicate and host checks refer to the refined catalogues.

- ▶ Eight $z = 0$ calls preserve all six particle arrays bit for bit, including sizes and counts.
- ▶ At $z = 0$, each tested fixed density field gives identical catalogues at 32 and 64 threads.
- ▶ Ordinary density recomputation changes up to 7 printed rows per replay; see the numerical appendix.

More haloes pass the mass cut

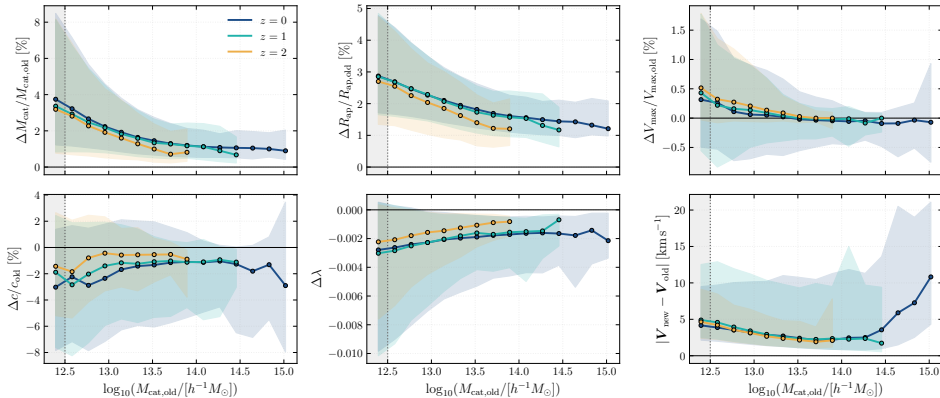


All published populations. Open low-count bins: $0 < N < 20$; ratios require both $N \geq 20$. Bars: $\sqrt{N}$ only.
Grey below $\log_{10} M = 12.5$: $z=0$ literature reference; no convergence claim for the revised finder.

Median mass and radius increase

Same-snapshot finder comparison; $N_p = 1024^3$, $L = 512 h^{-1}\text{Mpc}$

Finder revisions: a8c7715 to 8107a644e7a7



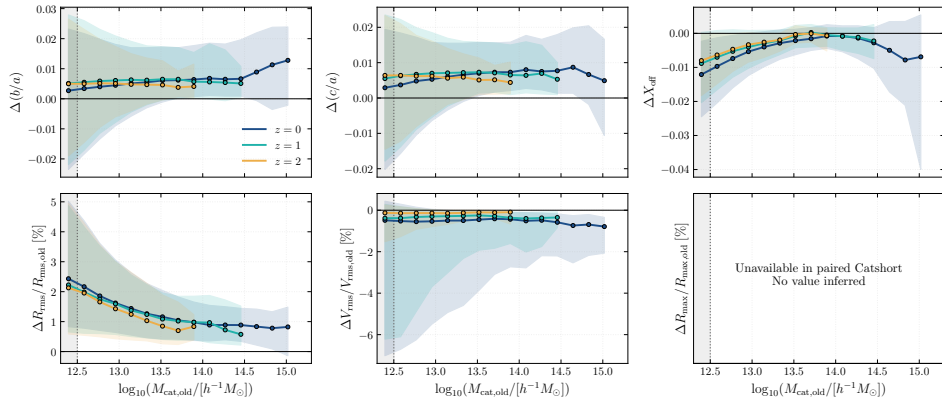
Conservative positional matches only. New minus old; median and 16-84 percentiles; $N_{\text{valid}} \geq 20$ per bin.

Grey below $\log_{10} M = 12.5$: $z=0$ literature reference; no convergence claim for the revised finder.

Internal structure responds to the refinement

Same-snapshot finder comparison; $N_p = 1024^3$, $L = 512 h^{-1} \text{Mpc}$

Finder revisions: a8c7715 to 8107a644e7a7

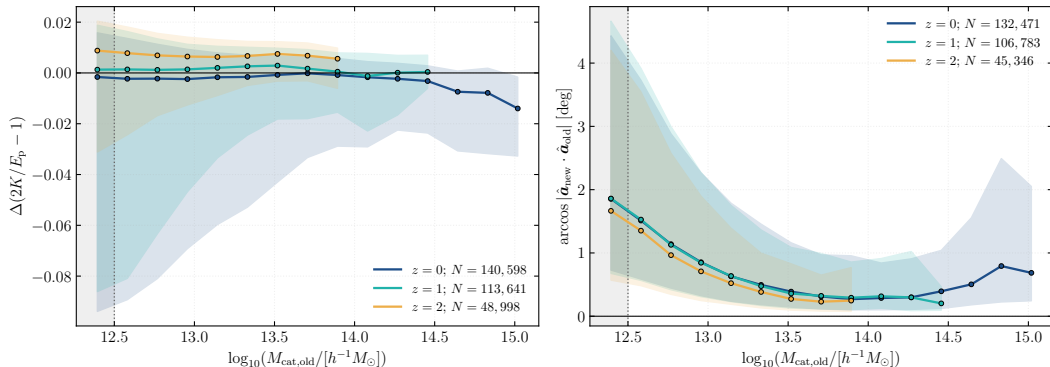


Conservative positional matches only. New minus old; median and 16-84 percentiles; $N_{\text{valid}} \geq 20$ per bin.

Grey below $\log_{10} M = 12.5$: $z = 0$ literature reference; no convergence claim for the revised finder.

Energy and major-axis directions change

Same-snapshot finder comparison; $N_p = 1024^3$, $L = 512 h^{-1} \text{Mpc}$
Finder revisions: a8c7715 to 8107a644e7a7



Directions: valid ratios and $\max(b/a, c/a) < 0.9$ in both catalogues; raw eigenvalue gaps unavailable.

Medians and 16–84 percentiles; $N_{\text{bin}} \geq 20$; bands are not errors on the median.

Grey below $\log_{10} M = 12.5$: $z=0$ literature reference; no convergence claim for the revised finder.

Abundance and halo properties change

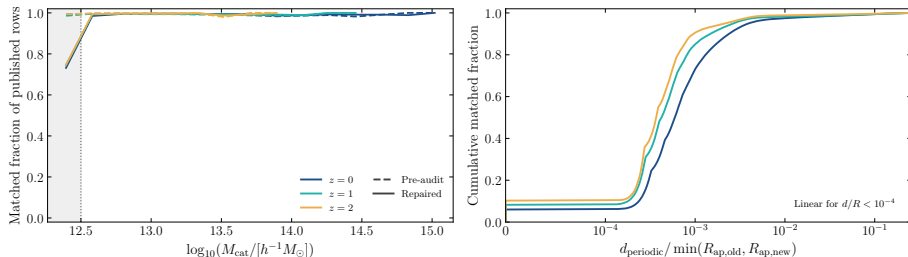
- ▶ At $z = 0$: 141,357 to 147,042 haloes (+4.0%). Several physics and configuration repairs contribute.
- ▶ For matched $z = 0$ haloes with both masses above $10^{12.5} h^{-1} M_{\odot}$, median changes are +2.2% in mass, +2.2% in radius and +0.07% in $V_{\max}$.
- ▶ No identical bound sets or host-exclusion violations survive in the checked refined catalogues.

Bound mass and internal properties share one converged population. Reported radius and total mass retain the empirical aperture extension. Precision applications need mass and force-resolution calibration.

Appendix: matching coverage

Same-snapshot finder comparison; $N_p = 1024^3$, $L = 512 h^{-1} \text{Mpc}$

Finder revisions: a8c7715 to 8107a644e7a7



	Total	Matched	Invalid	No target	Non-mutual	Too far	Ambiguous
Old, $z=0$	141,357	140,598	0	0	346	413	0
New, $z=0$	147,042	140,598	0	0	6,031	413	0
Old, $z=1$	114,454	113,641	0	0	444	368	1
New, $z=1$	119,688	113,641	0	0	5,678	368	1
Old, $z=2$	49,205	48,998	0	0	113	94	0
New, $z=2$	52,370	48,998	0	0	3,278	94	0

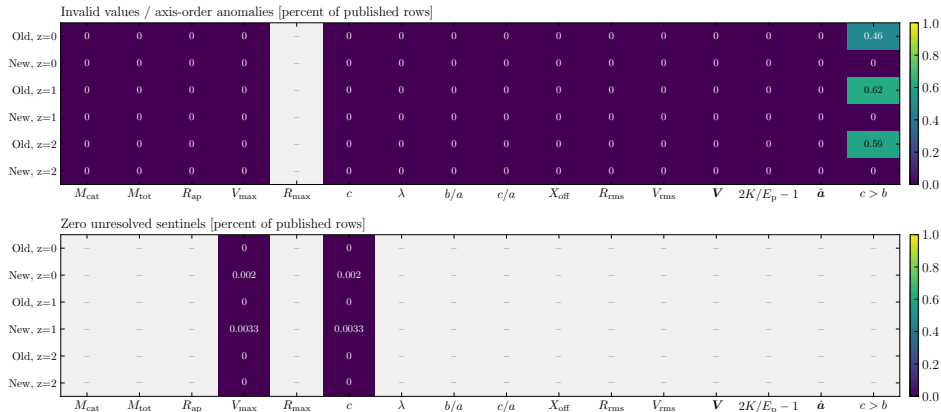
Table categories are exclusive and sum to the published row count. Unmatched = all categories except Matched.

Grey below $\log_{10} M = 12.5$: $z=0$ literature reference; no convergence claim for the revised finder.

Appendix: valid and unresolved values

Same-snapshot finder comparison; $N_p = 1024^3$, $L = 512 h^{-1} \text{Mpc}$

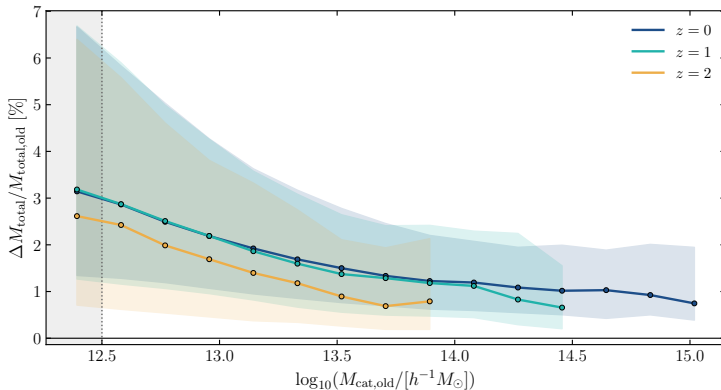
Finder revisions: a8c7715 to 8107a644e7a7



Grey: unavailable/not sentinel-defined. $c > b$: separate axis-order anomaly. Candidate rejections require finder logs.

Appendix: total aperture mass

Same-snapshot finder comparison; $N_p = 1024^3$, $L = 512 h^{-1}\text{Mpc}$
Finder revisions: a8c7715 to 8107a644e7a7



Aperture mass; matched objects only. Bands are halo-to-halo percentile ranges.
Grey below $\log_{10} M = 12.5$: $z = 0$ literature reference; no convergence claim for the revised finder.

Appendix: a concrete duplicate-survival case

Earlier 128^3 integration test: three close candidate pairs survived because their particle sets differed slightly.

Pair	Shared / smaller set	d	R_{host}
1	2109/2110	0.0035	1.216
2	326/329	0.0245	0.699
3	1441/1445	0.0155	1.082

Distances and radii are in h^{-1} Mpc.

Mass ties now use stable candidate order. A lower-priority centre inside its host is suppressed: all three extra entries disappear.

This example is separate from the new 1024^3 validation.

Appendix: density rounding sensitivity

- ▶ 512^3 pilot: 2/71,587 rows vary between ordinary 16/32-thread replays.
- ▶ Fixing the density field gives identical finder outputs. Changing only the field reproduces those two rows.
- ▶ 1024^3 : changing the density field changes 7 of 147,042 rows, including one bound-particle count. Each fixed-field 32/64-thread group is identical.

Shape follow-up: sort the two empirically corrected transverse lengths. The other properties and memberships are unchanged by that fix.

Appendix: computational cost

- ▶ Shared COSMA8 allocations. Simulation job: 58.0 min on 64 cores; batch peak 166.6 GiB, allocated 192 GiB.
- ▶ Standard $z = 0$ replay at 64 threads: pre-audit 46.0 s, refined 91.8 s ($2.00\times$). The refined finder performs additional SO and iterative-unbinding work.
- ▶ Controlled $32 \rightarrow 64$ threads: $1.62\times$ speedup (81% of ideal doubling); two repeats per thread count on one fixed field.
- ▶ Simulation, replay, fixed-field and plot jobs: 11948372, 11948467, 11948491, 11948494.

Appendix: definitions and reproducibility

- ▶ Pre-audit finder: a8c7715; refined finder: 8107a644e7a7.
- ▶ The old standalone baseline retains its first-call configuration defect; historical multi-output inline runs need their own interpretation.
- ▶ Rmax is absent from the historical catalogue. Concentration is a mass/radius/Vmax diagnostic; major directions are compared modulo sign.

Input, source, binary, snapshot and figure hashes, full status counts, matching exclusions and scripts accompany this deck.